Response to Reviewers

Manuscript: *Transfer Learning for Meta-analysis Under Covariate Shift*
Submission #86

General Response to All Reviewers and Chairs

We sincerely thank the Chairs and all reviewers for their careful reading of the manuscript and for the thoughtful, detailed, and constructive feedback. We are encouraged by the reviewers' recognition of the importance of the problem setting, the novelty of the placebo-anchored proxy–gold framework, and the potential applicability to real-world clinical and regulatory decision-making.

Across reviews, several common themes emerged, including requests for clearer articulation of assumptions, more explicit discussion of identification versus regularization, broader empirical validation, and improved clarity in presentation. In the revised manuscript, we plan to address these points by:

- clarifying the role of each assumption and explicitly distinguishing identification conditions from modeling or regularization choices;
- moderating theoretical claims where appropriate and emphasizing empirically testable sensitivity analyses;
- expanding experimental comparisons and robustness checks; and
- improving exposition, figures, and organization for accessibility to a broader audience.

Below, we provide a point-by-point response to each reviewer in blue coloured text.

1 Response to Reviewer dTHHH

Summary

This manuscript presents a novel placebo-anchored transfer learning framework for individual-patient-data meta-analysis in settings where covariate shift exists between randomized controlled trials (RCTs) and the target population, and where conventional network meta-analysis assumptions (e.g., shared comparator, network connectivity) break down.

Summary of Strengths

- The topic is timely and aligns well with growing interest in trial generalizability and causal machine learning.
- The core methodological idea, treating source IPD as proxy labels and target placebo outcomes as gold labels for anchored calibration, is well-structured.
- Embedding the correction into a doubly robust learner with known trial propensities is theoretically coherent and practical.
- Results demonstrate improved CATE estimation, baseline calibration, and transportability robustness relative to baseline models.

We thank the reviewer for their kind comments. We will now address the concerns raised.

Summary of Weaknesses

1. *Evaluation limited to synthetic data.* The framework is evaluated only on synthetic data and not on real RCT data.
   We understand the reviewer's concern regarding this. To address this issue we have included ...

2. *Assumption A5: low-dimensional structured covariate-driven shift.* Assumption A5 posits low-dimensional structured covariate-driven shift. In practice, site heterogeneity may arise from interactions, nonlinearities, or unobserved effect modifiers.

   [[ZilongW: @Ali can you find clinical trial papers / estimation papers that make utilize this assumption? This assumption is basically Hamsa's Proxy Gold paper's key assumption.]]

   We apologize for the lack of clarity in the original presentation and thank the reviewer for raising this important concern. We regret that Assumption A5 may have been interpreted as an identification assumption. We would like to clarify that Assumption A5 is *not* intended to guarantee full causal identification or transportability. Rather, it is a *regularity condition for estimation* that restricts the class of admissible cross-site deviations and enables calibrated extrapolation when treated outcomes are unavailable in the target population.

   Assumption A5 is motivated by both the transfer learning and clinical prediction literatures, which consistently find that cross-study heterogeneity is often driven by a limited number of clinically meaningful factors (e.g., baseline risk calibration, key biomarkers, disease severity, or measurement practices), rather than wholesale changes in the outcome-generating mechanism. In this sense, A5 reflects a parsimonious modeling assumption analogous to recalibration or limited coefficient revision in individual participant data meta-analysis and clinical model updating. We emphasize that A5 restricts the complexity of outcome-regression contrasts, not the covariate distributions themselves.

Importantly, A5 does not require transport bias to be exactly sparse or perfectly linear. Existing theory shows that LASSO-based corrections remain valid under approximate sparsity and degrade gracefully when the assumption is only partially satisfied (e.g., Bastani, 2021). We further emphasize that A5 concerns sparsity of *outcome-model contrasts* across sites, not sparsity of covariate shifts themselves.

Accordingly, A5 should be interpreted as a structural regularity assumption that improves estimation efficiency when approximately satisfied, rather than as a sharp identification condition. To address the reviewer's concern directly, we will include additional sensitivity analyses in the revision that examine nonlinear and non-sparse transport bias, to empirically characterize the scope and limitations of placebo anchoring.

3. *Limited evaluation metrics.* The evaluation focuses mainly on CATE and ATE RMSE. Other metrics may also need to be considered.

4. *Missing comparisons with existing methods.* Comparison with existing transportability or meta-analytic methods or benchmarks is missing.

2 Response to Reviewer `QwUG`

Summary

The manuscript is well written, clearly structured, and technically rigorous, with detailed experiments and robustness checks. The manuscript would benefit from more discussion of key assumptions in real-world clinical trial settings and clearer description of the experimental data sources.

Summary of Strengths

- The manuscript introduces a placebo-anchored proxy–gold transfer learning framework, which is an innovative way to combine abundant multi-site RCT data (proxy labels) with high-fidelity placebo outcomes from the gold labels.
- The manuscript is methodologically rigorous. Stage-wise anchoring using target placebo outcomes, and doubly robust orthogonalization using known randomization propensities, demonstrate careful research design.
- The manuscript includes strong experimental design. Well-designed evaluation metrics—Precision in Estimation of Heterogeneous Effects (PEHE), mean absolute error of the average treatment effect (ATE), and calibration RMSE for treated and placebo outcome regressions (μ_1 and μ_0)—capture both individual-level and population-level performance.
- Writing is clear and structure is logical.

Summary of Weaknesses

1. *Justification of Assumptions A4 and A5 in real-world settings.* Assumptions A4 and A5, specialized to the multi-site RCT setting, are crucial for performance. However, the manuscript does not sufficiently justify why real-world clinical trials or clinical data would satisfy these assumptions. Additional discussion is suggested.

   We thank the reviewer for this suggestion and agree that additional clinical context would strengthen the manuscript. We emphasize that Assumptions A4 and A5 are intended as *regularity conditions for stable estimation* in multi-site randomized trials, rather than strong biological invariance or causal identification assumptions.

   Assumption A4 has now been refined into a Lipschitz smoothness condition on outcome regressions: it requires that conditional mean outcomes vary continuously with baseline covariates within each site. This assumption rules out arbitrarily unstable responses to small covariate perturbations and ensures that moderate covariate shift across trial populations does not induce unbounded changes in predicted outcomes. Such smoothness assumptions are standard in nonparametric regression and causal learning, and are implicitly relied upon whenever outcome models are evaluated outside their exact training distribution. Importantly, A4 does not assert exchangeability, treatment-effect homogeneity, or transportability across sites.

   Assumption A5 concerns how outcome regressions differ across trial populations. Rather than assuming that treatment effects or baseline risks are identical across sites, A5 posits that systematic cross-site differences can be reasonably approximated by a low-complexity correction to a well-learned proxy model. Clinically, this reflects the common observation that between-trial heterogeneity is often driven by a limited number of interpretable factors—such as baseline risk calibration, disease severity, key biomarkers, or measurement and enrollment practices—rather than by wholesale changes in the outcome-generating mechanism.

This perspective is consistent with individual participant data meta-analysis and clinical prediction model updating, where cross-study heterogeneity is typically addressed through parsimonious recalibration or limited effect-modification, rather than unrestricted refitting of complex models. Importantly, Assumption A5 is *no longer* an identification assumption: it defines a working approximation class that enables stable estimation and calibrated extrapolation when treated outcomes are unavailable in the target population. When A5 holds approximately, estimation error decomposes into stochastic error and approximation error; when it is violated, performance degrades smoothly toward proxy-only baselines rather than exhibiting a sharp failure.

[[ZilongW: @Ali can you find clinical trial papers / estimation papers that make this claim? This assumption is basically Hamsa's Proxy Gold paper's key assumption.]]

2. *Description of experimental data sources.* The sources of the experimental data are not clearly described. It is recommended to more explicitly describe the data sources and data processing methods.

3 Response to Reviewer SDFf

Summary

The paper proposes a three-stage pipeline that (i) learns low-variance outcome models on pooled source RCT IPD (proxy labels), (ii) debiases the baseline using scarce target placebo (gold) outcomes via a sparse linear correction (LASSO), and (iii) embeds the anchored outcome models into a doubly robust (DR) / orthogonal learner to estimate patient-level CATEs. However, several of the core assumptions and statistical-rate claims appear underspecified or overly optimistic for the data regime under consideration (e.g., enough gold samples for cross-validation and estimation, bounded covariate shift without guaranteed bounded outcome regression, and strong assumptions on the transportability of a sparse covariate subset across sites). The proposed method may be promising in practice, but the theoretical conditions currently feel stronger than what the empirical setting can support.

The manuscript would be strengthened by additional experiments (e.g., non-linear treated-arm corrections, non-linear data-generating processes, comparisons against existing transport or meta-analytic methods), deeper evaluation of experiments via many Monte Carlo realizations (instead of a single synthetic data generation realization), and statistical hypothesis testing to more rigorously demonstrate performance gains in the proposed approach.

Summary of Strengths

1. The paper clearly articulates a practical and understudied problem of estimating treatment effects in a target population with only placebo outcomes available, using multi-trial RCT IPD as proxy data. The "proxy–gold" formalism provides a nice decomposition of the treatment effect into two terms: one term from the proxy outcomes (external trials) and one term from the gold outcomes (internal targeted trial).

2. The ablation studies clearly show the contribution of each stage: proxy-only, anchor-only, and proxy+anchor (full model), which demonstrate where improvements are driven from for CATE estimation.

3. The modular three-stage design enables flexible learners, where Stage 1 permits any high-capacity learner (e.g., random forest, neural network), while Stages 2 and 3 attempt to recover causal validity via calibration to target placebo and anchored treated outcomes.

4. The manuscript lays out comprehensive empirical robustness check concepts that illustrate practical failure modes and build intuition about where the method succeeds or fails.

Summary of Weaknesses

1. *Conditional exchangeability and unmeasured effect modifiers.* A core requirement in transport theory is conditional exchangeability $Y(0), Y(1) \perp C|X$ i.e., that outcomes are independent of the trial indicator C given covariates X. The paper explicitly states that this does not hold, as outcome models from source trials may be biased for the target and covariate distributions drift across sites. This raises concerns that covariate shift may be correlated with unmeasured effect modifiers (e.g., sites recruiting more severe patients may also differ in unmeasured clinical protocols). It is unclear mathematically whether the sparse bias regression term is sufficient to address this issue.
   [[ZilongW: @Ali, look at the multi-site RCT papers you sent me and how they phrased the

identification assumptions. In particular, look out for the multi-site indicators (Dahabreh 2023, Hong 2025, Rott 2024)]]

We thank the reviewer for highlighting this fundamental issue and fully agree with the concern. We emphasize at the outset that our method does not rely on, nor does it claim to recover, classical conditional exchangeability across sites. In particular, we do not assume

$$(Y(0), Y(1) \perp C|X)$$

nor do we claim nonparametric identification of target treatment effects in the presence of unmeasured effect modifiers.

To clarify the scope of our contribution, it is useful to distinguish baseline risk transportability from treatment-effect transportability. Existing multi-site transport and meta-analytic methods typically require assumptions such as:

- Exchangeability over sites (Dahabreh et al., 2023):

$$Y(a) \perp C|X,$$

- Sample ignorability for treatment effects (Hong et al., 2025):

$$C \perp (Y(1) - Y(0))|X.$$

Both of which rule out unmeasured effect modifiers. As noted by Rott et al. (2025), these assumptions are often restrictive and implausible in practice.

Our approach explicitly departs from this identification paradigm. Rather than assuming transport bias is zero, we treat it as present but structured. Specifically, we leverage observed target-site placebo outcomes to anchor and recalibrate baseline risk. This anchoring step operates entirely through $Y(0)$ and therefore addresses baseline risk miscalibration, not unobserved treatment-effect heterogeneity.

Importantly, we do not claim that placebo anchoring recovers bias arising from unmeasured effect modifiers that differentially affect treated and untreated outcomes. If such modifiers induce treatment-specific deviations beyond baseline risk, they remain fundamentally unidentifiable without treated outcomes in the target population. Our estimator should therefore be interpreted as a calibrated, model-based transport estimator, not a design-based identification result.

Assumption A5 formalizes this distinction. It is introduced as a regularity condition on the complexity of cross-site deviations in the outcome regression, not as a substitute for conditional exchangeability. When A5 holds approximately, anchoring improves calibration and estimation stability; when it is violated, performance degrades smoothly toward proxy-only baselines rather than exhibiting a sharp identification failure (see response to Q5 further below regarding concerns about A5).

The manuscript has been revised to make this point explicit [[ZilongW: and to avoid confusion We will revise the manuscript to make this separation explicit and to avoid language suggesting that unmeasured effect modification is resolved. Instead, we emphasize that our contribution lies in robust calibration under acknowledged transport bias, rather than causal identification under strong exchangeability assumptions.]]

2. *Total variation bound and outcome stability.* The argument that $\mu_a(x)$ is the same between sites because the total variation (TV) distance of covariates distributions across trials (sites) is bounded is questionable. Also, the total variation between two probability density functions is always less than or equal to 1, so why is it bounded by infinity in A4? Translating TV distance bounds into bounds on the predicted outcome of μ_a requires further smoothness or Lipschitz assumptions on $\mu_a(x)$, which are not discussed. For example (with a different distribution measure), if μ_a is Lipschitz with constant L, then

$$W_1(\mu_\sharp P, \mu_\sharp Q) \leq L W_1(P, Q),$$

where W_1 is the 1-Wasserstein metric and $\mu_\sharp$ is the pushforward operation on the input covariate distribution (Villani et al., 2008).

We thank the reviewer for carefully pointing out the issues with our original formulation of Assumption A4. We agree that the original statement was imprecise and potentially misleading, and we apologize for the sloppiness in its presentation.

In the original manuscript, Assumption A4 was stated as a bound on covariate shift across sites using total variation distance:

$$\textbf{(Original A4)} \qquad d_{\mathrm{TV}}\big(P_{X|c_1}, P_{X|c_2}\big) \leq \Delta < \infty,$$

for all sites c_1, c_2, where d_{TV} denotes total variation distance.

As the reviewer correctly notes, total variation distance is always bounded above by 1, and therefore the condition $\Delta < \infty$ is redundant. More importantly, as pointed out by the reviewer, bounding covariate shift in total variation does not, by itself, imply stability of outcome regressions or bounded prediction error without additional smoothness assumptions. We further clarify that we never intended to assume or claim that $\mu_a(x)$ is identical across sites; rather, the original assumption was meant to impose weak regularity for estimation, not equality or transportability.

To address this concern, we have revised Assumption A4 to a Lipschitz-type regularity condition on the conditional outcome regressions, removing total variation distance entirely. The revised assumption is stated as follows:

$$\textbf{(Revised A4)} \qquad \big|\mu_{a,c}(x) - \mu_{a,c}(x')\big| \;\leq\; L\,\|x - x'\|, \quad \forall a \in \{0,1\},\ \forall c,\ \forall x, x',$$

where $\mu_{a,c}(x) = \mathbb{E}[Y \mid A = a, X = x, c]$ and $L < \infty$ is a uniform Lipschitz constant.

This revised assumption serves solely as a *regularity condition* ensuring that moderate covariate shift across sites does not induce arbitrarily large changes in predicted outcomes for nearby covariate profiles. It does *not* imply exchangeability, transportability, or stability of treatment effects across sites, and it is not used to justify causal identification.

All structural assumptions regarding how cross-site differences manifest in outcome regressions are now explicitly delegated to Assumption A5, which we frame as a regularity condition enabling calibrated estimation rather than as an identification assumption. We believe this revision clarifies the role of Assumption A4 and directly addresses the reviewer's concern.

3. *Monte Carlo evaluation and statistical testing.* For the synthetic experiments, there should be many Monte Carlo realizations to compute means and standard deviations of PEHE, MAE of ATE, calibration RMSE for μ_0 and μ_1 in both ablation and robustness checks. Otherwise, it

is unclear whether observed improvements are due to randomness in data generation. Statistical hypothesis testing (e.g., Friedman test, with causal variants as treatments and metrics as blocks) is suggested. The causal variants in table II are the "treatments" and the metrics are the "blocks". Lastly you could vary the number of covariates used.

We agree with the reviewer and [[ZilongW: add experiments]]

4. *Claim of estimation without a treated arm in the target.* There is a claim that treatment effects are estimated in the target population with no treated arm (first page in the introduction), but if so, how is $\beta_{1,0}$ estimated? You may transport the support but not the estimated coefficients. Later in the paper, it is stated that both arms are present in the gold data.

   We thank the reviewer for pointing out this ambiguity. We agree that the original wording could be interpreted as claiming identification or direct estimation of target treated-arm coefficients from target data, which is not what our method does. We have revised the manuscript to make this distinction explicit.

   No nonparametric identification from target placebo alone. When the target trial has no treated outcomes, $\mu_{1,0}(x)$ and $\tau_0(x) = \mu_{1,0}(x) - \mu_{0,0}(x)$ are not nonparametrically identified from target data alone. We make no claim to the contrary.

   Updated procedure in disconnected targets (Option B). We now define a *working-model* treated regression using a cross-arm transfer operator learned from source sites where both arms are observed. Concretely:

   - We fit proxy regressions $\widehat{\mu}_a^{\text{proxy}}$ on pooled sources (Stage 1).
   - We estimate the target placebo correction $\widehat{\beta}_{0,0}$ via sparse regression on target placebo residuals (Stage 2).
   - We estimate source-site sparse bias coefficients $\widehat{\beta}_{0,c}, \widehat{\beta}_{1,c}$ and learn a rank-r matrix $\widehat{M}$ by reduced-rank regression minimizing $\|\widehat{B}_1 - \widehat{M}\widehat{B}_0\|_F^2$ (new Assumption A6 / procedure).
   - We then *construct* the target treated correction as $\widehat{\beta}_{1,0} := \widehat{M}\widehat{\beta}_{0,0}$ and define $\widehat{\mu}_{1,0}^{\text{anch}}(x) = \widehat{\mu}_1^{\text{proxy}}(x) + x^\top \widehat{\beta}_{1,0}$.

   This is a data-driven extrapolation based on cross-site information, not an identification result from target data.

   Resolution of the "both arms in gold" confusion. In some experiments we include target treated outcomes for evaluation/benchmarking and for Option A (when they exist). We now clearly separate (i) the motivating disconnected-target regime (no treated outcomes used for estimation) from (ii) evaluation settings where treated outcomes are observed solely to compute error metrics. We revised the introduction and methods sections to reflect this distinction.

5. *Restrictive linear and sparse bias model.* The model constrains the bias term to be linear and sparse, which is restrictive and not well justified theoretically. If the true correction term were nonlinear or non-sparse, it is unclear how the method would perform. Sensitivity analyses are suggested for violations of Assumption A5 (non-sparse or nonlinear bias).

   We apologize for the lack of clarity in the original presentation and thank the reviewer for highlighting an important practical concern. In particular, we regret that Assumption A5 was stated to be an identification assumption. We would like to clarify that Assumption A5 is *not* intended to guarantee full causal identification or transportability of treatment effects. Rather, it is a *structural regularity condition* that restricts the class of admissible cross-site deviations and enables stable estimation and calibrated extrapolation when treated outcomes are unavailable in the target population.

Assumption A5 is motivated by a growing body of transfer learning and clinical prediction literature showing that differences between study populations are often driven by a limited number of clinically meaningful factors, rather than by wholesale changes in the outcome-generating mechanism. In practice, site- or population-level differences in randomized trials frequently arise from shifts in baseline risk calibration, prevalence of key biomarkers, disease severity distributions, or measurement and enrollment practices, rather than from entirely different treatment-response mechanisms.

In the proxy–gold transfer sparse regression based learning literature, Bastani (2021) formalizes this idea by modeling discrepancies between abundant proxy data and scarce gold-standard data as low-dimensional corrections to an otherwise well-learned prediction model. Importantly, this assumption is supported empirically in healthcare prediction tasks, where sparse adjustments are often sufficient to reconcile differences between external datasets and target clinical populations.

Crucially, Assumption A5 does *not* require that transport bias be exactly sparse or perfectly linear. As discussed in Bastani (2021), the bias term recovered in the second stage of a two-step transfer estimator typically consists of a dominant low-dimensional signal combined with residual noise propagated from the proxy model. Established high-dimensional theory shows that LASSO-based procedures remain statistically valid under *approximate sparsity*, with estimation error decomposing into a variance component from the proxy stage and a regularization error term. Bühlmann and Van De Geer (2011); Belloni et al. (2012); Bastani (2021) further show that this structure yields meaningful sample-efficiency gains while degrading gracefully when sparsity is only approximate.

Importantly, approximate sparsity does not restore identification of the true target CATE. When Assumption A5 is violated, the estimator converges to a different working-model target and incurs a non-vanishing approximation bias, which we explicitly characterize in our theory and robustness experiments.

From a clinical standpoint, this assumption aligns closely with standard practices in individual participant data meta-analysis and model transportability. These literatures emphasize parsimonious modeling of effect modification across trials, often focusing on a small number of known or suspected modifiers, rather than unrestricted interaction structures (Riley and Fisher, 2021; Dahabreh et al., 2023). Similarly, the clinical prediction model updating literature consistently finds that cross-site differences are most effectively addressed through recalibration or limited coefficient revision, rather than full refitting of complex models (Wynants et al., 2017; Van Calster et al., 2019; Steyerberg, 2019).

This sparsity-of-contrast perspective has also been formalized in modern multi-source transfer learning theory. Li et al. (2022) study settings with multiple auxiliary datasets and explicitly assume that informative sources differ from the target through sparse contrast vectors, while allowing non-informative or heterogeneous sources to coexist. They establish minimax-optimal rates and robustness guarantees under this assumption, directly reflecting the realities of multi-center clinical trial data. Tian and Feng (2023) extend these results to high-dimensional generalized linear models, showing that sparse contrasts are sufficient for improved prediction and estimation, while violations can lead to negative transfer—an observation that mirrors clinical experience when inappropriate external data are over-weighted.

Finally, we emphasize that Assumption A5 does *not* assert that covariate distributions themselves shift sparsely across sites. That is, we do not assume for sites c, c'

$$\|X_c - X_{c'}\|_0 \le q \quad \text{for small } q \in \mathbb{N}.$$

Rather, the assumption concerns the *outcome regression*, namely that differences in how patient characteristics map to outcomes across sites can be well approximated by sparse contrasts in regression coefficients. Under a linear approximation,

$$Y_{c_i} = \beta_c^\top X_c + \varepsilon_c, \qquad Y_{c_j} = \beta_{c'}^\top X_{c'} + \varepsilon_{c'},$$

we assume

$$\|\beta_c - \beta_{c'}\|_0 \le q \quad \text{for small } q \in \mathbb{N}.$$

This reflects the clinical belief that only a small subset of patient features—such as key risk factors or effect modifiers—drive systematic differences in outcomes across trial populations.

Accordingly, Assumption A5 is now stated as a regularity condition for estimation rather than an identification assumption. When the assumption holds approximately, placebo anchoring provides informative calibration signal and improves estimation efficiency; when it is violated, performance degrades gracefully toward proxy-only baselines rather than exhibiting a sharp identification failure.

6. *Shared support across treatment arms.* It is unclear why it is necessarily true that different treatment arms share the same support set of covariates.

   [[ZilongW: @Ali, while I think this assumption made is reasonable, let me know if you find papers where this is not the case. I don't think this is a big deal but let's make this reviewer happy.]]

   We thank the reviewer for raising this point, as it highlights an important distinction between biological assumptions and design-based assumptions. Shared support across treatment arms is not a biological claim about treatment response, but a standard *design property* of randomized controlled trials.

   In our setting, treatment assignment occurs after enrollment and is randomized with known probabilities. Consequently, for any baseline covariate pattern observed at enrollment, each participant has a positive probability of being assigned to either treatment arm. This corresponds to the standard positivity (or overlap) condition in causal inference for randomized experiments, under which baseline covariates have overlapping support across treatment groups by design (Imbens and Rubin, 2015; Hernán and Robins, 2010; Dahabreh et al., 2023)

   We emphasize that this assumption applies only to *baseline covariates measured prior to treatment assignment*. It does not require that post-treatment variables, intermediate outcomes, or biologically driven response mechanisms share support across arms. Nor does it assert that treatment effects are homogeneous across covariate values. Rather, it ensures that conditional expectations such as $\mathbb{E}[Y \mid A = a, X = x]$ are well-defined for both arms over the observed covariate space.

   We also acknowledge that violations of this assumption can occur in practice, for example due to deterministic treatment assignment, protocol-driven exclusions after enrollment, or conditioning on post-baseline variables. Such violations would indeed invalidate the overlap condition and, consequently, the proposed approach. We now explicitly clarify this distinction, scope, and failure mode in the revised manuscript.

7. *Disconnected trial networks.* The paper motivates disconnected trial networks, but does not explicitly evaluate performance on disconnected networks (e.g., no A–B or B–C comparisons, only A–C gaps). For the synthetic dataset, it is unclear how disconnected networks are constructed if estimation requires treated arms. Clarification is needed.

We thank the reviewer for noting that the original manuscript motivated *disconnected trial networks* but did not explicitly evaluate performance in a setting where standard network meta-analysis (NMA) or IPD-NMA is infeasible by design. We agree with this concern and have revised the manuscript to both clarify the definition of disconnectedness and include an explicit experimental evaluation.

Clarification of disconnected trial networks. We now define disconnectedness in the standard NMA sense. Let $\mathcal{T}$ denote the set of treatments (including placebo), and define a treatment graph whose nodes correspond to treatments and whose edges represent direct randomized comparisons observed in trials. A network is disconnected if the target treatment is not connected to the target population through a chain of shared comparators. In such cases, conventional NMA/IPD-NMA estimators are undefined or rely on additional, typically untestable, assumptions.

Explicit disconnected-network experiment. We added a synthetic multi-treatment experiment in which the treatment graph is disconnected by construction. Specifically, we consider treatments

$$\mathcal{T} = \{0, A, B\},$$

where 0 denotes placebo. The data-generating process is defined as follows.

- *Source trials.* Two source sites are observed:

  $$\begin{aligned} &\text{Site } c = 1: \quad A \in \{0, A\}, \\ &\text{Site } c = 2: \quad A \in \{0, B\}, \end{aligned}$$

  with randomized assignment within each site. No site contains a direct comparison between A and B.
- *Target site.* The target site $c = 0$ contains only placebo observations:

  $$\mathcal{D}_{\text{gold}} = \{(X_i^{(0)}, A_i^{(0)} = 0, Y_i^{(0)})\}_{i=1}^{m},$$

  and no treated outcomes for either A or B.
- *Outcome model.* Outcomes are generated according to

  $$Y = \mu_0(X) + \mathbb{I}(A = A)\,\tau_A(X) + \mathbb{I}(A = B)\,\tau_B(X) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2),$$

  where $\mu_0(X)$ is the baseline risk function and $\tau_A(X), \tau_B(X)$ denote heterogeneous treatment effects. Covariate distributions differ across sites to induce covariate shift.

In this setup, the treatment graph is disconnected for the target estimand $\mathbb{E}[\tau_A(X) \mid c = 0]$. Standard NMA/IPD-NMA methods cannot be applied because there is no path connecting treatment A to the target population through shared comparator arms under the usual comparability assumptions.

Evaluation and findings. We evaluated (i) Proxy-Only, (ii) Anchor-Only, and (iii) the full placebo-anchored DR-Learner in this disconnected setting. As expected, conventional NMA/IPD-NMA baselines are undefined and therefore omitted by design. The proposed method remains well-defined because treated outcomes from source trials serve as proxy information, while target placebo outcomes anchor baseline risk. Empirically, the results mirror the connected-network ablations: placebo anchoring improves target calibration, and orthogonalization stabilizes CATE estimation under covariate shift.

[[ZilongW: Add results here]]

Interpretation. We emphasize that this experiment does not claim design-based identification of treatment effects from the target data alone. Rather, it demonstrates that the proposed framework continues to yield calibrated, model-based estimates in settings where conventional network-based estimators fail due to disconnection. This directly operationalizes the motivating scenario described in the introduction and clarifies the scope of applicability of the method.

8. *Efficiency and finite-sample claims.* In the discussion of the joint placebo-anchored estimator, it seems unlikely that the estimator is correctly specified given the linear bias term. As a result, claims of efficiency may not hold. It is also unclear what sample size is referred to in $\sqrt{n}$ rates (proxy samples, gold samples, or total samples). Given the small gold-sample regime, finite-sample bounds describing how estimation error depends on sample size would be helpful.

We thank the reviewer for this important and technically substantive comment. We agree that our earlier phrasing around "$\sqrt{n}$" rates and, in particular, "efficiency" was not sufficiently precise in the proxy–gold, multi-site setting, especially given the multi-stage correction and the Step B cross-arm transfer used when the target has no treated outcomes. We have therefore revised both the theoretical framing and the language in the manuscript. Below we (i) clarify the estimand, (ii) specify the asymptotic regime and what "$\sqrt{n}$" refers to, (iii) give a concise proof sketch for asymptotic linearity and normality that explicitly accounts for Step B and the updated Assumption A6, and (iv) provide an explicit finite-sample error decomposition that separates sampling, nuisance, and structural approximation terms.

Clarification of the estimand (working-model CATE). Let $S \in \{0, 1, \dots, C\}$ index sites, with $S = 0$ the target. The true target CATE is

$$\tau_0(x) := \mathbb{E}[Y(1) - Y(0) \mid X = x, S = 0] = \mu_{1,0}(x) - \mu_{0,0}(x).$$

In disconnected regimes (no treated outcomes in the target), $\mu_{1,0}$ and hence $\tau_0(x)$ are not nonparametrically identified. Accordingly, our asymptotic results concern a *working-model* target

$$\tau^\star(x) := \mu^\star_{1,0}(x) - \mu_{0,0}(x),$$

where $\mu^\star_{1,0}$ is the population limit induced by Stage 1 proxy learning, Stage 2 gold calibration on target placebo, and Step B cross-arm transfer under Assumption A6. Assumption A5 defines a stable low-complexity class for cross-site deviations, yielding a well-posed working-model limit; violations appear as explicit structural approximation terms.

Asymptotic regime and meaning of "$\sqrt{n}$. We observe $C+1$ randomized trials with site-specific sample sizes $\{n_c\}_{c=0}^{C}$ and total Stage 3 sample size

$$N := \sum_{c=0}^{C} n_c, \qquad \pi_c := n_c/N.$$

We work in a fixed-C triangular-array regime where $n_c \to \infty$ for each site and $\pi_c \to \pi_c^\star$. All "$\sqrt{n}$" statements refer to $\sqrt{N}$. Because sites have different covariate and outcome distributions, the asymptotic variance is a site-weighted mixture rather than an i.i.d. variance.

Which sample sizes govern which parts of the error. Although the leading sampling term is governed by N, the nuisance remainder depends on *distinct* effective sample sizes: Stage 1 proxy regressions are learned from pooled source data with size $n_{\text{proxy}} := \sum_{c=1}^{C} n_c$, Stage 2 calibration uses only target placebo (gold) observations with size $m_0 := |\{i : S_i = 0, A_i = 0\}|$ (and m_1 if target treated outcomes are used under Option A), and Step B estimates the cross-arm transfer operator (or coupling parameters) using source sites, with an effective size that scales with the amount of source treated-and-placebo information (e.g., $\sum_{c=1}^{C} n_c$ in the simplest pooled implementation, or $\sum_{c=1}^{C} 1$ if the operator is estimated from site-level summaries). As a result, finite-sample behavior in the small-gold regime is typically dominated by m_0, even when N and n_{proxy} are large, and is additionally sensitive to the accuracy of Step B when the target has no treated outcomes.

Proof sketch (asymptotic linearity and CLT for $\tau^\star(x)$). Fix x.

Step 1: Orthogonal score construction. Using known propensities from randomization (Assumption A2) and overlap (Assumption A3), Stage 3 forms a cross-fitted doubly robust pseudo-outcome

$$\psi_i = \widehat{\tau}^{\text{anch}}(X_i) + \frac{A_i - e_{S_i}(X_i)}{e_{S_i}(X_i)\{1 - e_{S_i}(X_i)\}} \Big\{ Y_i - \widehat{\mu}^{\text{anch}}_{A_i,S_i}(X_i) \Big\},$$

and estimates $\tau^\star(x)$ by regressing ψ_i on X_i. Under Option B, the target treated regression inside $\widehat{\tau}^{\text{anch}}$ is constructed via Step B (e.g., $\widehat{\mu}^{\text{anch}}_{1,0}(x) = \widehat{\mu}^{\text{proxy}}_{1}(x) + \widehat{\delta}_{1,0}(x)$ with $\widehat{\delta}_{1,0}$ obtained by transferring the placebo correction through the Step B operator). The corresponding population score is Neyman-orthogonal: first-order perturbations in nuisance functions do not affect the target moment, so nuisance estimation error enters only at second order.

Step 2: Asymptotic linear expansion. Let $\mu^\star_{a,s}$ denote the population limits of the anchored regressions (including the Step B-induced limit for $\mu^\star_{1,0}$ under Option B) and define the oracle influence function

$$\phi(O; x) := \frac{A - e_S}{e_S(1 - e_S)} \Big\{ Y - \mu^\star_{A,S}(X) \Big\} + \tau^\star(X) - \tau^\star(x).$$

Standard orthogonalization arguments with cross-fitting yield the pointwise expansion

$$\widehat{\tau}(x) - \tau^\star(x) = \frac{1}{N} \sum_{i=1}^{N} \phi(O_i; x) + R_N(x),$$

where $R_N(x)$ is a second-order remainder.

Step 3: Remainder control and role of Assumptions A4–A6 (including Step B). Write nuisance errors as: (i) proxy prediction error δ_{proxy} (learned from n_{proxy} and evaluated on the target distribution), (ii) placebo calibration error δ_B (Stage 2, learned from m_0), and (iii) Step B transfer error δ_M (error in the cross-arm transfer operator/coupling learned from sources). Orthogonality implies a product-rate remainder of the form

$$R_N(x) = O_p\Big(\delta_{\text{proxy}}\big(\delta_B + \delta_M\big)\Big) + \text{Approx}(x).$$

Assumption A4 is a regularity condition ensuring stability of the regression functionals under moderate covariate shift, so that proxy errors remain controlled when evaluated on the target covariate distribution. Assumption A5 ensures that Stage 2 targets a low-complexity correction and separates estimation error from approximation residuals. Assumption A6 (together with the Step B construction) determines the *working-model* mapping from placebo corrections to treated corrections when treated target outcomes are unavailable; violations contribute a structural (non-vanishing) approximation term.

Step 4: Asymptotic normality. Provided $R_N(x) = o_p(N^{-1/2})$, a triangular-array central limit theorem yields

$$\sqrt{N}\big(\widehat{\tau}(x) - \tau^\star(x)\big) \ \Rightarrow\ \mathcal{N}\left(0,\ \sum_{c=0}^{C} \pi_c^\star \operatorname{Var}\big(\phi_c(\cdot\,; x)\big)\right).$$

This step relies only on independence, finite second moments, and stabilization of site proportions, conditional on the validity of the asymptotic linear expansion.

Finite-sample error decomposition. For each fixed x, the estimation error admits the bound

$$|\widehat{\tau}(x) - \tau^\star(x)| \ \le\ \underbrace{O_p(N^{-1/2})}_{\text{sampling fluctuation}} \ +\ \underbrace{C\,\delta_{\text{proxy}}\big(\delta_B + \delta_M\big)}_{\text{orthogonal remainder (nuisance)}} \ +\ \underbrace{\text{Approx}_{\text{A5–A6}}(x)}_{\text{structural approximation error}},$$

where the structural term collects misspecification of the low-complexity bias class (A5) and cross-arm nontransfer not captured by the Step B coupling (A6). In the sparse-linear instantiation, a concrete representation is

$$\text{Approx}_{\text{A5–A6}}(x) = |r_{0,0}(x)| + |x^\top \nu_0| + |r_{1,0}(x)|,$$

where $r_{0,0}$ is the within-arm approximation residual for the target placebo correction (A5), $r_{1,0}$ is the within-arm approximation residual for the target treated regression (A5), and $x^\top \nu_0$ is the target-specific cross-arm nontransfer component in A6 (i.e., the part of the treated correction not predictable from the placebo correction via the Step B transfer operator). When Option A is used (some treated target outcomes), the role of $x^\top \nu_0$ is partially attenuated because $\mu_{1,0}$ is directly estimated on its observed support. In the sparse-linear calibration example, δ_B scales as $\sqrt{s \log p / m_0}$, highlighting the importance of the gold placebo size in small-m_0 regimes; δ_M scales with the source information used to learn the transfer operator in Step B.

Implications and revisions. We therefore agree with the reviewer on two key points and have revised the manuscript accordingly. First, we no longer claim semiparametric efficiency for the nonparametrically unidentified target CATE $\tau_0(x)$. Second, all "$\sqrt{n}$" statements are now explicitly tied to the total Stage 3 sample size $N = \sum_c n_c$, with finite-sample behavior governed separately by proxy sample size, target placebo sample size, Step B transfer-operator error, and structural approximation error arising from Assumptions A5–A6.

Revised theoretical statement. Under Assumptions A4–A6, cross-fitting, and standard DML regularity conditions, the proposed estimator is Neyman-orthogonal and asymptotically linear for the working-model CATE $\tau^\star(x)$ at each fixed covariate value x. Its first-order behavior is governed by a site-weighted influence function, while finite-sample error depends explicitly on proxy sample size, target placebo sample size, Step B transfer-operator accuracy, and approximation error induced by violations of Assumptions A5–A6. No unconditional claim of nonparametric identification or semiparametric efficiency for $\tau_0(x)$ is made.

9. *LASSO scaling and cross-validation.* LASSO estimates depend on covariate scaling, but the manuscript does not describe how covariates are standardized or how this affects interpretability of the estimated coefficients. It is also unclear how cross-validation is performed to select the regularization parameter given the small number of gold-standard labels, which may lead to noisy selection.

   We thank the reviewer for highlighting these important implementation details. We agree that both covariate scaling and regularization selection are critical for the stability and interpretability of the sparse calibration step, particularly in the small-gold regime.

   Covariate scaling and interpretability. In Stage 2, the sparse bias correction is estimated via a LASSO regression of gold-standard placebo outcomes on covariates (or learned representations). Prior to fitting the LASSO, all covariates entering the calibration regression are standardized *within the gold sample* to have zero mean and unit variance:

$$\tilde{X}_{ij} = \frac{X_{ij} - \bar{X}_j}{\hat{\sigma}_j}, \qquad \bar{X}_j = \frac{1}{m}\sum_{i=1}^{m} X_{ij}, \quad \hat{\sigma}_j^2 = \frac{1}{m}\sum_{i=1}^{m}(X_{ij} - \bar{X}_j)^2,$$

   where m denotes the number of gold (target placebo) observations. This standardization ensures that the ℓ_1 penalty acts uniformly across covariates and that variable selection is not driven by scale artifacts.

   Under this scaling, the estimated coefficients $\hat{\beta}_j$ correspond to the effect of a one-standard-deviation change in covariate X_j on the outcome scale. When reporting coefficients for interpretability, we back-transform to the original scale via $\hat{\beta}_j^{\text{orig}} = \hat{\beta}_j / \hat{\sigma}_j$ and adjust the intercept accordingly. We now state this explicitly in the manuscript.

   Regularization path and step sizes. We compute the LASSO solution along a standard logarithmic regularization path

$$\lambda \in \{\lambda_{\max}, \lambda_{\max}\alpha, \lambda_{\max}\alpha^2, \ldots\},$$

   where

$$\lambda_{\max} = \left\| \frac{1}{m}\tilde{X}^\top y \right\|_\infty$$

   is the smallest value yielding the all-zero solution, and $\alpha \in [0.9, 0.95]$ is the multiplicative step size. In our experiments we use $\alpha = 0.9$, resulting in a path of 50–100 candidate values depending on the problem dimension. This follows standard practice in high-dimensional regression and is consistent with common implementations (e.g., `glmnet`).

Cross-validation with limited gold labels. Because the number of gold-standard placebo observations is small, naive K-fold cross-validation can be unstable. To mitigate this, we adopt the following procedure:

(a) We use repeated K-fold cross-validation with $K \in \{3, 5\}$, chosen so that each validation fold contains a nontrivial number of gold observations.

(b) For each split, we evaluate prediction error on held-out gold outcomes using squared error.

(c) We select the regularization parameter using the *one-standard-error (1-SE) rule*, choosing the largest λ whose validation error is within one standard error of the minimum. This favors sparser and more stable solutions in small-sample regimes.

(d) As a robustness check, we also report sensitivity analyses using a fixed λ proportional to $\sqrt{\log p/m}$, which is the theoretically motivated scaling for sparse linear models.

This procedure substantially reduces variance in λ selection while preserving the intended bias–variance tradeoff of the anchoring step.

Impact on downstream estimation. Finally, we emphasize that the Stage 3 estimator is Neyman-orthogonal with respect to the nuisance parameters, including the LASSO-estimated calibration coefficients. As a result, moderate variability in $\hat{\beta}$ induced by small-gold cross-validation affects the final CATE estimator only at second order, a point we now highlight explicitly in the revision.

Revision summary. We have updated the manuscript to (a) explicitly describe covariate standardization and coefficient back-transformation, (b) state the regularization path and step sizes used for LASSO, and (c) clarify how cross-validation is performed and stabilized in the small-gold regime.

10. *Missing comparisons with existing methods.* The paper does not compare against other statistical approaches for covariate shift or transportability mentioned in the related work. Including such baselines would strengthen the paper, especially against methods that do not rely on the same modeling assumptions.

Response to Q10 (Missing comparisons with existing methods). We thank the reviewer for this constructive suggestion and agree that including additional baselines would strengthen the empirical evaluation. We have therefore expanded the comparison set to include representative statistical approaches for covariate shift, transportability, and multi-trial evidence synthesis that differ in their modeling assumptions from the proposed placebo-anchored framework.

Added baseline categories. The additional methods fall into four complementary categories:

(a) **Reweighting-based transport estimators.** We include inverse probability weighting (IPW) and stabilized IPW estimators that reweight source trial participants to match the target covariate distribution (Dahabreh et al., 2023; Westreich et al., 2017). These methods rely on correct specification of the site-assignment model and covariate overlap, and provide a natural contrast to our outcome-model-based anchoring approach.

(b) **Outcome-regression transport estimators.** We compare against outcome-regression approaches that pool source trials and fit a single outcome model with site indicators and site–covariate interactions (Hong et al., 2025). Predictions are then evaluated on the target covariate distribution. This baseline captures classical regression-based transport under effect-modification assumptions.

(c) **Augmented and doubly robust transport estimators.** We include augmented IPW (AIPW) and doubly robust transport estimators that combine outcome regression with site reweighting (Dahabreh et al., 2019). These methods provide robustness to misspecification of either the outcome model or the site model, but still require conditional exchangeability across sites.

(d) **IPD meta-analytic baselines.** We include individual-participant-data meta-analysis (IPD-MA) baselines, including fixed-effects and random-effects models with treatment–covariate interactions (Riley and Fisher, 2021). These methods represent standard practice in multi-trial synthesis and highlight differences between our proxy–gold calibration approach and hierarchical pooling strategies.

Baselines tailored to the target-data regime. Because some transport estimators require treated outcomes in the target population, we report two evaluation settings:

- *Oracle setting:* treated outcomes are available in the target site for evaluation, allowing all baselines to be computed.
- *Restricted setting:* only target placebo outcomes are used for estimation, in which case baselines that require treated target outcomes are included only for reference and are clearly labeled as infeasible in practice.

This separation avoids unfairly penalizing methods whose data requirements differ from our motivating scenario.

Implementation details. All added baselines are implemented using standard generalized linear models, linear or penalized regressions, and reweighting estimators, following the specifications in the cited literature. Hyperparameters are selected via cross-validation where applicable, and all methods share the same feature set to ensure a fair comparison.

What these comparisons illustrate. These additional baselines allow us to demonstrate that:

- reweighting-based methods are sensitive to covariate overlap and degrade under strong covariate shift;
- regression-based and IPD-MA approaches perform well when exchangeability holds but can be miscalibrated when baseline risk differs across sites;
- the proposed placebo-anchored estimator remains competitive in calibration and CATE accuracy in regimes where treated target outcomes are unavailable or scarce.

Revision summary. We have added these baselines to the experimental section and clarified the data requirements and assumptions of each method, thereby positioning the proposed approach relative to established transport and meta-analytic alternatives that rely on different modeling assumptions. [[ZilongW: Add experiments and results]]

11. *Analytical degradation with cross-arm correlation.* Is it possible to add structural assumptions or identification results under Assumption A5 to quantify how CATE error degrades as cross-arm correlation decreases, analytically rather than only empirically?

    We thank the reviewer for this insightful suggestion. We agree that it is valuable to complement empirical "ρ-sweeps" with an analytical characterization of how error degrades as placebo-to-treated transport validity weakens. Importantly, however, **Assumption A5 alone is insufficient** to parameterize cross-arm degradation: A5 constrains *cross-site* deviations within each arm through a low-complexity class $\mathcal{D}$, but it does not relate the *treated-arm* and *placebo-arm* deviations in the target. Consequently, to quantify degradation as cross-arm correlation decreases, one must impose an additional *cross-arm structural condition* (or adopt an explicit sensitivity parameterization).

    In the revised manuscript, we formalize this coupling via **Assumption A6** and align it with our **Step B** construction. This yields an explicit analytical degradation statement that appears as the *structural approximation/bias* term in our updated finite-sample error decomposition (Appendix). Throughout, we emphasize that Neyman orthogonality yields robustness and a $\sqrt{N}$-CLT for a *working-model* target CATE $\tau^\star(x)$, but it does not imply semiparametric efficiency for the *true* target CATE $\tau_0(x)$, which is not nonparametrically identified in disconnected regimes.

    What can and cannot be done analytically in disconnected targets. In disconnected regimes (no treated outcomes in the target), the treated regression $\mu_{1,0}(x)$ (and hence the true target CATE $\tau_0(x)$) is not nonparametrically identified from target data. Consequently, any analytical "degradation" statement necessarily concerns the *structural bias* induced by the working restriction used to extrapolate treated outcomes, rather than an identification result for $\tau_0(x)$ under A5 alone.

    Recap of A5 and why it does not determine cross-arm degradation. Let $S = 0$ denote the target site and $S = c \geq 1$ source sites, and define

    $$\mu_{a,c}(x) := \mathbb{E}[Y \mid A = a, X = x, S = c], \qquad a \in \{0, 1\}.$$

    Stage 1 learns proxy regressions μ_a^{proxy} from sources. Under Assumption A5, for each arm a and site c,

    $$\mu_{a,c}(x) = \mu_a^{\text{proxy}}(x) + \delta_{a,c}(x) + r_{a,c}(x), \qquad \delta_{a,c} \in \mathcal{D},$$

    where $r_{a,c}$ denotes the approximation residual when A5 holds only approximately. Assumption A5 controls *within-arm* deviations across sites, but it does not relate $\delta_{1,0}$ to $\delta_{0,0}$. Therefore, A5 alone cannot quantify how placebo anchoring transfers to treated outcomes in the target.

    Cross-arm coupling (A6) aligned with Step B: operator transfer and nontransfer component. To parameterize cross-arm transport validity in the same form used by Step B, we use the sparse linear instantiation $\delta_{a,c}(x) = x^\top \beta_{a,c}$ and impose Assumption A6: there exists a low-rank linear operator $M^\star \in \mathbb{R}^{p \times p}$ such that for each site c,

    $$\beta_{1,c} = M^\star \beta_{0,c} + \nu_c, \qquad \operatorname{rank}(M^\star) \leq r,$$

    where ν_c captures treatment-specific components not explained by placebo-to-treated transfer. Under Option B, Step B estimates $M^\star$ from source sites to obtain $\widehat{M}$ and constructs the target treated-arm correction as $\widehat{\beta}_{1,0} = \widehat{M}\,\widehat{\beta}_{0,0}$. Decreasing "cross-arm correlation" corresponds to larger nontransfer signal (larger ν_0) and/or greater mismatch between the true treated deviation and its operator-transferred component.

Analytical degradation bound: structural bias relative to the true target CATE. Our Stage 3 orthogonal learner delivers $\sqrt{N}$-rate inference for the *working-model* target

$$\tau^\star(x) := \mu^\star_{1,0}(x) - \mu_{0,0}(x),$$

where $\mu^\star_{1,0}(x) = \mu_1^{\text{proxy}}(x) + x^\top \beta^\star_{1,0}$ with $\beta^\star_{1,0} := M^\star \beta^\dagger_{0,0}$ (the best-approximation coefficient $\beta^\dagger_{0,0}$ for the target placebo deviation in $\mathcal{D}$). The total error relative to the true target CATE decomposes as

$$|\widehat{\tau}(x) - \tau_0(x)| \le |\widehat{\tau}(x) - \tau^\star(x)| + |\tau^\star(x) - \tau_0(x)|.$$

The first term is the statistical error controlled by our updated appendix, while the second term is the *structural degradation* due to imperfect cross-arm transfer. Using Assumption A6 and allowing A5 to hold only approximately on the treated arm,

$$\tau^\star(x) - \tau_0(x) = \mu^\star_{1,0}(x) - \mu_{1,0}(x) = x^\top \nu_0 + r_{1,0}(x),$$

which yields the explicit pointwise degradation bound

$$|\tau^\star(x) - \tau_0(x)| \le |x^\top \nu_0| + |r_{1,0}(x)|.$$

Taking $L_2(P_0)$ norms gives

$$\|\tau^\star - \tau_0\|_{L_2(P_0)} \le \|x^\top \nu_0\|_{L_2(P_0)} + \|r_{1,0}\|_{L_2(P_0)}.$$

Thus, *as cross-arm transport validity decreases*, degradation increases through an interpretable nontransfer component (ν_0) and any within-arm approximation residual for the treated arm ($r_{1,0}$).

Connection to the updated finite-sample decomposition: separation of statistical and structural terms. The Appendix proves a $\sqrt{N}$-CLT for $\widehat{\tau}(x)$ around $\tau^\star(x)$ and the finite-sample bound

$$\|\widehat{\tau} - \tau^\star\|_{L_2(P_0)} \lesssim O_p(N^{-1/2}) + \delta_{\text{proxy}}\big(\delta_B + \delta_M\big),$$

where δ_{proxy} is Stage 1 proxy error, δ_B is target-placebo calibration error, and δ_M is Step B transfer-operator estimation error. Combining this with the structural bound yields the transparent total decomposition

$$\|\widehat{\tau} - \tau_0\|_{L_2(P_0)} \le \underbrace{O_p(N^{-1/2}) + \delta_{\text{proxy}}(\delta_B + \delta_M)}_{\text{sampling + orthogonal nuisance (working-model)}} + \underbrace{\|x^\top \nu_0\|_{L_2(P_0)} + \|r_{1,0}\|_{L_2(P_0)}}_{\text{cross-arm structural degradation}}.$$

This directly answers the reviewer's request: we now provide an *analytical* characterization of how error degrades as placebo-to-treated transfer weakens (via ν_0 and $r_{1,0}$), and we clarify that this degradation is fundamentally a *structural* component distinct from the $\sqrt{N}$ stochastic fluctuation controlled by orthogonality.

Remark (relation to scalar ρ sweeps). A scalar ρ-sweep corresponds to a special case of A6 with $M^\star = \rho I$, so that $\beta_{1,0} = \rho\beta_{0,0} + \nu_0$. Our operator-based formulation strictly generalizes this while retaining an explicit degradation bound and matching the Step B procedure used in the revision.

12. *Site imbalance and error propagation.* How does the model perform when there is a large imbalance between sites in terms of sample size? Additionally, errors and uncertainty may accumulate due to multiple estimation stages that depend on previous stages, unlike hierarchical or multilevel modeling approaches. Further discussion of error propagation is suggested.

    We thank the reviewer for raising two closely related practical concerns: (i) performance under strong *site sample-size imbalance*, and (ii) potential *error propagation* across the three estimation stages.

    (i) Site imbalance. Let n_c denote the sample size of site c and $N = \sum_{c=0}^{C} n_c$. Our Stage 3 estimator admits an influence-function expansion of the form

$$\hat{\tau}(x) - \tau^{\star}(x) = \frac{1}{N} \sum_{c=0}^{C} \sum_{i=1}^{n_c} \phi_c(O_{ci}; x) \; + \; R_N(x),$$

    so the leading variance contribution is naturally *site-weighted* by $\pi_c := n_c/N$. Consequently, severe imbalance affects the estimator primarily through (a) the variance mixture

$$\sum_{c=0}^{C} \pi_c \, \mathrm{Var}\big(\phi_c(\cdot; x)\big),$$

    and (b) the *effective gold budget* $m := n_0^{(A=0)}$ governing how well baseline risk is anchored at the target site.

    Practically, this yields two informative regimes:

    - *Small gold (target placebo) but large proxy:* anchoring uncertainty dominates, so performance is most sensitive to m and to the (approximate) sparsity/linearity of the correction in Stage 2.
    - *Large gold but small proxy:* proxy nuisance error dominates, so gains from anchoring may persist but are limited by the accuracy of Stage 1 estimates of μ_a^{proxy}.

    To address this concern empirically, we add a *site-imbalance sweep* in the synthetic study: we fix total N and vary the imbalance ratio $n_{\max}/n_{\min}$, and separately vary m holding the proxy sample sizes fixed. We report mean and dispersion (over Monte Carlo replications) for PEHE, ATE error, and calibration RMSE, thereby quantifying degradation as imbalance grows and isolating whether the bottleneck is proxy learning or gold anchoring.

    (ii) Error propagation across stages. We agree that multi-stage procedures can accumulate error. Our key mitigation is that Stage 3 uses a Neyman-orthogonal / doubly robust score with cross-fitting, which renders the final estimator *first-order insensitive* to nuisance estimation error. Concretely, the second-order remainder admits the standard product structure By orthogonality and cross-fitting, the second-order remainder admits a product structure

$$R_N(x) = O_p\Big(\|\hat{\eta} - \eta^{\star}\|_{\mathcal{N},1} \cdot \|\hat{\eta} - \eta^{\star}\|_{\mathcal{N},2}\Big) \; + \; \mathrm{Approx}(x),$$

    where η collects the Stage 1 proxy regressions and Stage 2 calibration parameters, $\|\cdot\|_{\mathcal{N},1}, \|\cdot\|_{\mathcal{N},2}$ are appropriate L_2-type norms, and $\mathrm{Approx}(x)$ denotes structural approximation error arising from Assumption A5 (and, when relevant, cross-arm coupling in Assumption A6).

Since Stage 2 calibration is estimated from the target placebo sample of size m_0, while Stage 1 proxy regressions are estimated from the pooled source sample, a sufficient condition for asymptotic linearity is the standard product-rate requirement

$$\|\hat{\eta} - \eta^\star\|_{\mathcal{N},1} \cdot \|\hat{\eta} - \eta^\star\|_{\mathcal{N},2} = o_p(N^{-1/2}),$$

which holds under mild rate conditions when $m_0 \to \infty$ and the proxy sample grows faster than $N^{1/2}$.

Uncertainty quantification and relation to hierarchical modeling. We further clarify that hierarchical/multilevel models can be appealing under explicit cross-site exchangeability/random-effects assumptions, but they typically target a different estimand and assumption set than our placebo-anchored setting. For uncertainty quantification, we add a practical variance estimate based on the empirical variance of the estimated influence-function contributions (with the natural site weighting through π_c), and we include a sensitivity check via fold-based re-fitting (and, in experiments, a bootstrap robustness check) to illustrate stability under small-gold regimes.

Revision summary. In the revision we (a) add explicit imbalance experiments, (b) make the finite-sample error decomposition explicit in terms of proxy nuisance error, gold anchoring error, and A5 approximation error, and (c) emphasize that orthogonality makes stagewise error propagation second-order rather than additive.

4 Response to Reviewer U94o

Summary

This paper addresses the problem of transportability of treatment effects from Randomized Controlled Trials (RCTs) to external, heterogeneous patient populations. In particular, it focuses on scenarios where the target treatment is different than those tested in the trial environment.

The authors attempt to solve this problem using a custom framework for training and then refining a doubly robust metalearner. First, a learner is trained on the available clinical trial data at all the sites that are available for a given therapy. Second, the placebo trials at a target site are used as a "gold" standard for training a LASSO regressor that is used to modify the prediction of the original regressor when the treatment is not provided.

These methods were validated using a simulated multi-center RCT dataset, constructed to simulate a covariate shift across sites.

Summary of Strengths

- The main strength of this paper lies in the customized doubly robust algorithm for integrating placebo patients from the site of interest into the predictor.
- The manuscript provides an in-depth analysis of bias and corresponding corrections that arise when dealing with heterogeneous treatment effects.

Summary of Weaknesses

1. *Assumption of placebo trials as a "gold standard."* The major weakness of the paper lies in the assumption that placebo trials constitute a "gold standard." The factors causing baseline risk to shift between source and target populations may not be the same factors, or act with the same magnitude, as those causing the treatment effect to shift.

   We thank the reviewer for raising this important conceptual concern. We clarify that our use of the term *proxy–gold* follows established terminology in the high-dimensional transfer learning literature and was *not* intended imply that placebo outcomes identify treatment effects. We apologize for this potential confusion. We definitely agree that placebo outcomes should *not* be interpreted as a causal or identification "gold standard" for treated outcomes or treatment effects in the target population.

   In particular, the proxy–gold terminology originates in sparse and high-dimensional transfer learning settings, where abundant but biased data sources ("proxies") are corrected using scarce but high-fidelity labels ("gold") that are reliable for a *specific component* of the prediction problem (Bastani, 2021; Li et al., 2022; Tian and Feng, 2023). In this literature, "gold" refers to labels that are unbiased for the target distribution of interest—not to a causal ground truth for all downstream quantities. Our usage follows this convention.

   In our setting, placebo outcomes are "gold" only in the sense that they provide an unbiased and site-specific calibration signal for the target population's *baseline outcome regression* $\mu_{0,0}(x)$ under randomization. They are not assumed to identify treated outcomes, treatment effects, or treatment-effect heterogeneity in the target population.

   Crucially, we do *not* assume that the mechanisms driving baseline risk shift and treatment-effect shift coincide. This distinction is central to our framework and is now formalized in the revised manuscript. Placebo anchoring corrects baseline miscalibration induced by covariate shift, while extrapolation to the treated arm is governed by an explicit structural regularity condition (Assumption A5), which defines a *working-model target* rather than a nonparametrically identified estimand.

To avoid further confusion, we [[ZilongW: have revised]] the manuscript to consistently emphasize that "proxy–gold" is a transfer-learning construct and to replace informal references to "gold standard" with "gold calibration signal" or "anchor," making clear that placebo data improve baseline alignment but do not, by themselves, guarantee correct transport of treatment effects.

2. *Exclusive reliance on synthetic data.* The paper only utilizes synthetic data, not de-identified or real-world data. The number of covariates used (three causal features and two confounders) is not representative of real scenarios, which typically involve dozens of different features.

3. *Insufficient description of data generation.* The manuscript does not describe the data generation process in sufficient detail. It is unclear what distributions were used for the covariates themselves, what weights were used for calculating the outcomes, and what sample sizes were used.

4. *Writing quality and presentation issues.* The paper is not well written and contains a number of grammatical and syntactic issues. For example, Table I is entirely unnecessary since it merely repeats information already in the text.

   Similarly, the section titled *Implementation* ("The simulation is implemented in Python with fixed random seeds to ensure reproducibility.") does not provide any meaningful contribution to the manuscript. Figures 2, 3, 4, and 5 are simply bar charts showing the same data already presented in Table II.

5 Response to Reviewer `hBhv`

Summary

A placebo-anchored transport framework is proposed to address the challenges in representing broader patient populations of randomized clinical trials.

Summary of Strengths

- The research problem and its significance are clearly articulated.
- An experiment is conducted to analyze the strength of the proposed methods, and the analysis is clearly presented.

Summary of Weaknesses

1. *Motivation relative to existing literature.* A literature review is presented and the weaknesses of existing methods are discussed. However, the motivation for the proposed method is not clearly articulated in terms of how it directly addresses the identified shortcomings.

   [[ZilongW: @Ali no need to rewrite yet. Maybe feed the intro into an LLM along with this comment and ask what can be changed. Don't rewrite anything yet.]]

2. *Lack of benchmarking against state-of-the-art methods.* The experiments only present comparisons among different settings of the proposed method. There is no benchmarking against state-of-the-art methods in the literature, making it unclear whether the proposed approach outperforms existing techniques, despite the analysis being interesting.

3. *Missing conclusion section.* A conclusion section is missing that would highlight the main contributions and findings of the proposed work.

6 Conclusion

We again thank the reviewers and the editorial team for their time and valuable feedback. We believe that addressing these comments will significantly strengthen the manuscript, and we appreciate the opportunity to revise and improve our work.

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